MARK SCHEME · TEACHER COPY

10BR · Chapter 10: Boolean Logic

Cambridge IGCSE Computer Science 0478

Paper 2 Algorithms, Programming and Logic

Total marks: 75

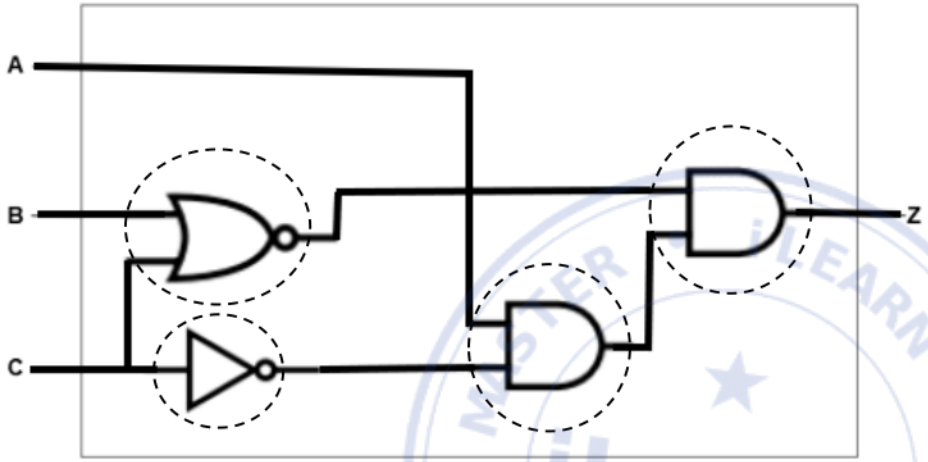
Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 75 marks

Question 1 · May/June 2023 Paper 23, original Q8 (a),(b) [8 marks]

Question	Answer	Marks
1(a)	**One** mark for each correct gate, with the correct input(s) as shown. 	4

Total: 75 marks

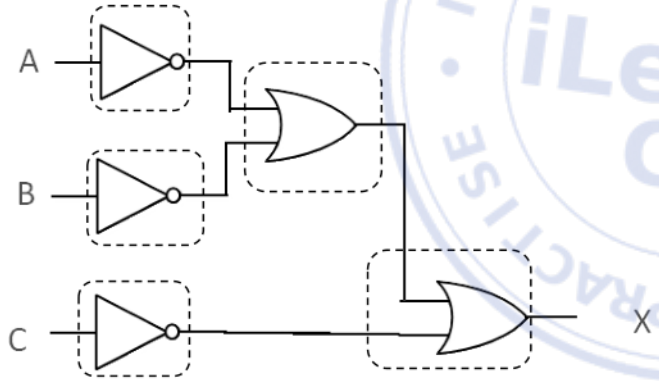
1(b)	**Four** marks for eight correct outputs. **Three** marks for six or seven correct outputs. **Two** marks for four or five correct outputs. **One** mark for two or three correct outputs	**4**

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

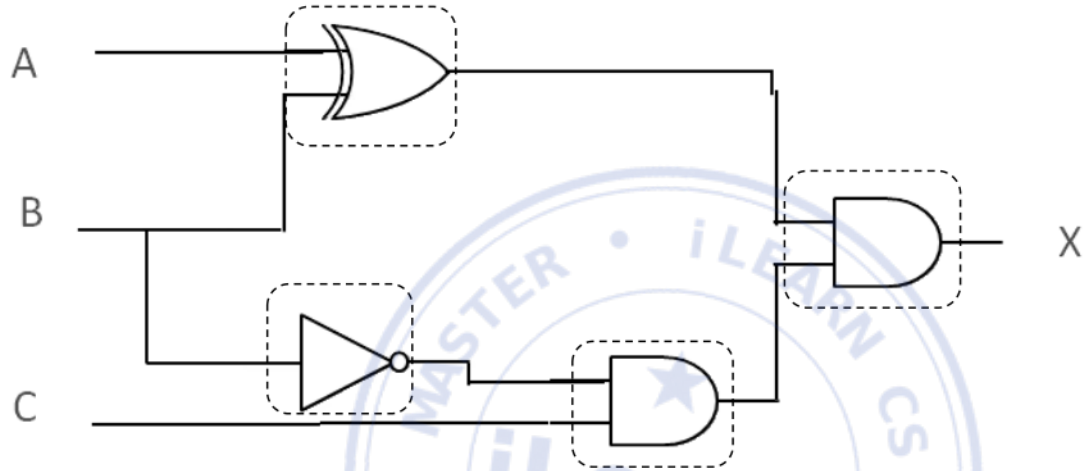
Question 2 · Oct/Nov 2023 Paper 23, original Q6 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks

2(a)	**One** mark for correct gate and **one** mark for correct truth table **AND** **A** **B** **X** 0 0 0 0 1 0 1 0 0 1 1 1	**2**
2(b)	**One** mark for correct gate and **one** mark for correct truth table **XOR // EOR** **A** **B** **X** 0 0 0 0 1 1 1 0 1 1 1 0	**2**

2(c)	**One** mark for correct gate and **one** mark for correct truth table **NOR** **A** | **B** | **X** 0 | 0 | 1 0 | 1 | 0 1 | 0 | 0 1 | 1 | 0	**2**
2(d)	**One** mark for each correct gate, with the correct input(s) as shown. 	**5**

Question 3 · Feb/March 2024 Paper 22, original Q7 (a),(b) [8 marks]

Question	Answer	Marks
3(a)	**One** mark for each correct gate, with the correct input(s) as shown. 	4

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Total: 75 marks

3(b)

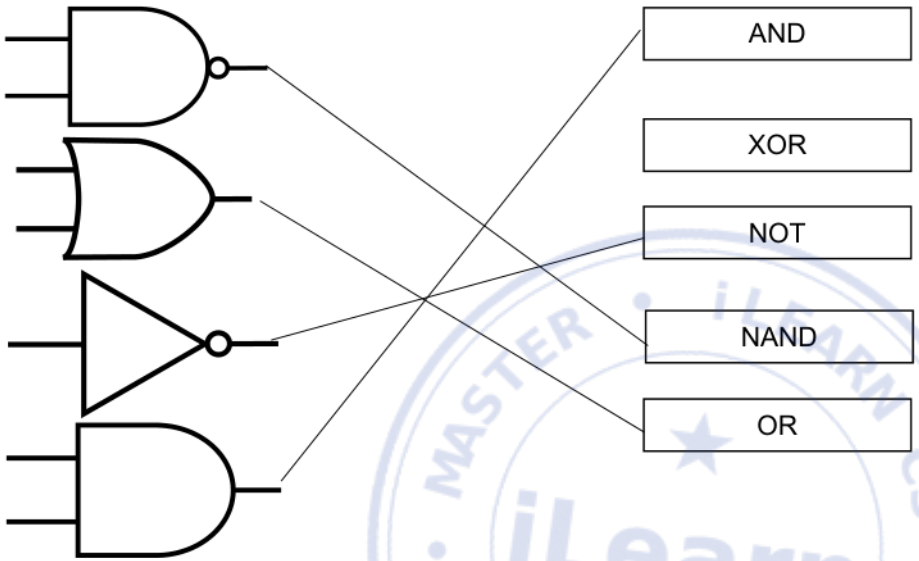
A	B	C	X
0	0	0	**0**
0	0	1	**0**
0	1	0	**0**
0	1	1	**0**
1	0	0	**0**
1	0	1	**1**
1	1	0	**0**
1	1	1	**0**

4 marks for 8 correct outputs
3 marks for 6/7 correct outputs
2 marks for 4/5 correct outputs
1 mark for 2/3 correct outputs

4

Question 4 · May/June 2024 Paper 23, original Q2 (a),(b) [8 marks]

Question	Answer	Marks

4(a)	**One** mark for each correct line **Logic gate symbol** **Logic function**  	**4**

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Total: 75 marks

Question	Answer	Marks
4(b)	<table><tr><th>A</th><th>B</th><th>C</th><th>Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table> **Four** marks for eight correct outputs. **Three** marks for six or seven correct outputs. **Two** marks for four or five correct outputs. **One** mark for two or three correct outputs	4

Question 5 · Oct/Nov 2024 Paper 21, original Q7 (a),(b) [7 marks]

Question	Answer	Marks
5(a)	**One** mark for each point • NOT **X** • **Y** XOR **Z** • expression correct: NOT **X** OR (**Y** XOR **Z**) // (**Y** XOR **Z**) OR NOT **X**	3

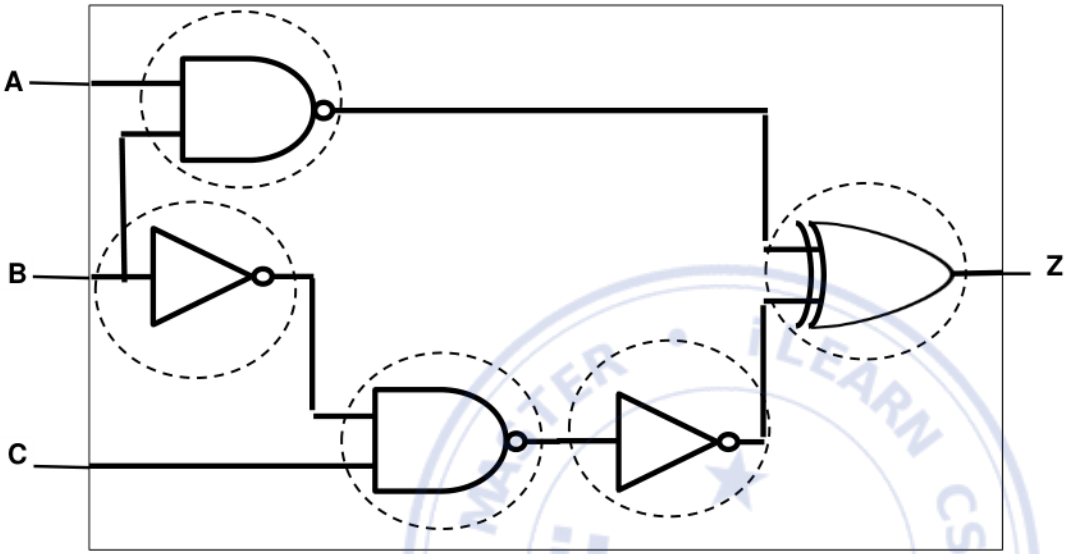
5(b)	4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs	**4**

X	Y	Z	W
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Question 6 · Feb/March 2025 Paper 22, original Q9 (a),(b) [9 marks]

Question	Answer	Marks

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6(a)	**One** mark for each correct gate, with the correct input(s) as shown. 	**5**

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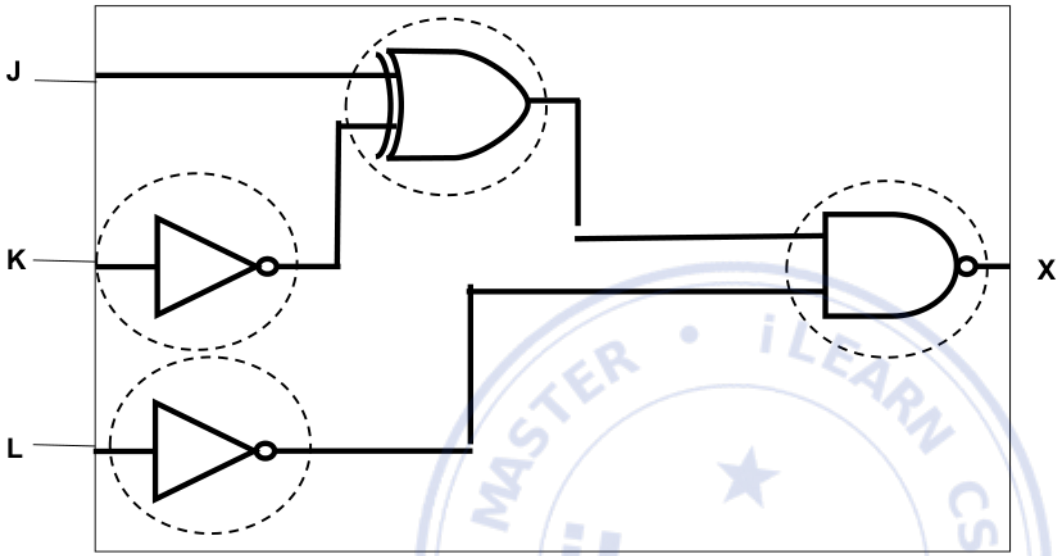
Total: 75 marks

6(b)	**Four** marks for all eight correct outputs **Three** marks for six or seven correct outputs **Two** marks for four or five correct outputs **One** mark for two or three correct outputs	**4**

A	B	C	Z
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

Question 7 · Oct/Nov 2025 Paper 22, original Q6 (a),(b) [8 marks]

Question	Answer	Marks

7(a)	**One** mark for each correct gate, with the correct input(s) as shown. 	**4**

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Total: 75 marks

7(b)	**Four** marks for eight correct outputs **Three** marks for six or seven correct outputs **Two** marks for four or five correct outputs **One** mark for two or three correct outputs	**4**

J	K	L	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Question 8 · May/June 2026 Paper 21, original Q8 (a),(b) [7 marks]

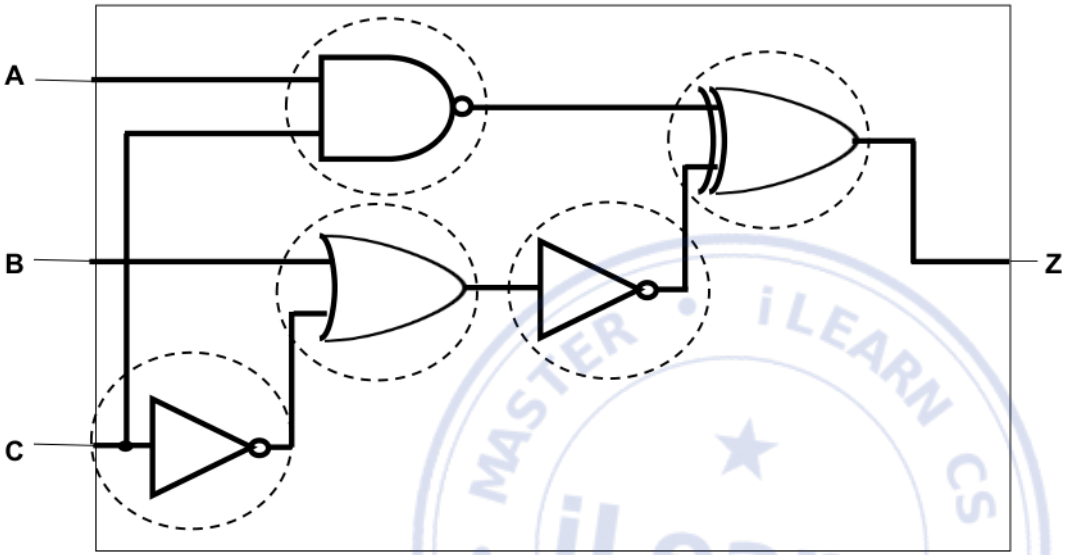
Question	Answer	Marks	Guidance
8(a)	**One** mark for each mark point: • A XOR B • B AND NOT C • correct expression: **X =** (A XOR B) **OR** (B AND NOT C) // **X =** A XOR B **OR** B AND NOT C	**3**	Allow FT if either MP1 or MP2 incorrect for MP3

8(b)	<table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table> 4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs	**4**	

Question 9 · May/June 2026 Paper 22, original Q8 (a),(b) [9 marks]

Question	Answer	Marks	Guidance

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9(a)	**One** mark for each correct gate, with the correct inputs as shown. 	**5**	

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9(b)	**Four** marks for eight correct outputs **Three** marks for six or seven correct outputs **Two** marks for four or five correct outputs **One** mark for two or three correct outputs	**4**	emboldened parts given in question

A	B	C	Z
0	**0**	**0**	1
0	**0**	**1**	0
0	**1**	**0**	1
0	**1**	**1**	1
1	**0**	**0**	1
1	**0**	**1**	1
1	**1**	**0**	1
1	**1**	**1**	0

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